

WASP-11b

R-0911

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_250909 · created 2026-07-29T06:11:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460927.8299 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.02 +/- 0.052
Transit depth (Rp/Rs)^2	0.039 +/- 0.2 [%]
Orbital Inclination (inc)	84.24 +/- 1.76
Airmass coefficient 1 (a1)	1.2 +/- 0.72
Airmass coefficient 2 (a2)	-0.21 +/- 0.4
Scatter in the residuals of the lightcurve fit is	54.18 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	6.5
Transit Duration (day)	0.019 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.02 ± 0.052 vs 0.131 (deviation 2.13σ)
Duration fit vs published	0.019 d vs 0.1075 d (deviation 3.05σ)
Mid-time O–C	-7.9 min
Depth SNR	0.4
Residual scatter	54.18 %

Light curve

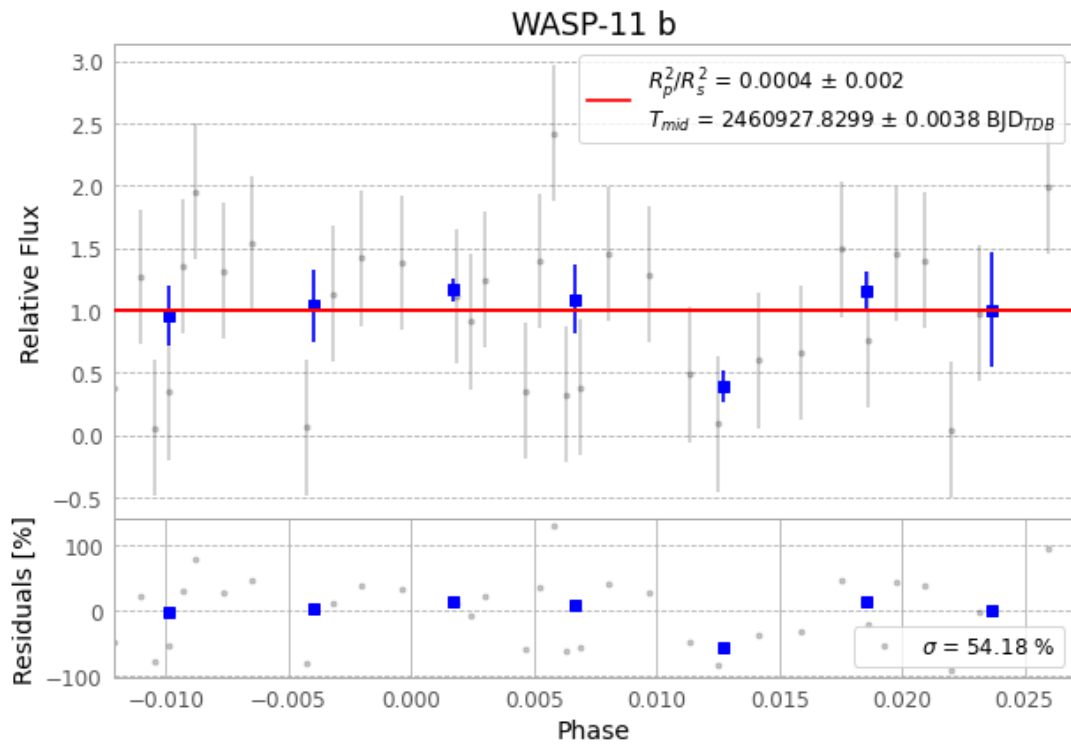


Figure 1 — FinalLightCurve_WASP-11 b_2025-09-08.png (SHA-256 1e066c145f7dd2cc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [494, 329], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3ee12f3c76d676f0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

73 FITS frames (48 MB), HATP-10250909063916.FITS → HATP-10250909101816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-250909.log (532971b35d9c...), FinalLightCurve_WASP-11 b_2025-09-08.png (1e066c145f7d...), FinalLightCurve_WASP-11 b_2025-09-08.csv (25a35f8a6602...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 06:11Z.